Special Issue on Fighting Fake: Disinformation and Misinformation

Fighting Misinformation on Social Media: Why People Share Fake News and Whether Credibility Indicators Help (Supplementary)

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Appendix A: Participants demographics

The sample consisted of 272 participants and reflects a broadly diverse demographic profile across ethnicity, age, and educational background. In terms of ethnicity, the largest groups were White (47%) and Asian/Pacific Islander (40%), with additional representation from Black or African American (7%), Hispanic or Latino (4%), and American Indian or Native American (2%). This distribution indicates that while certain groups were more predominant, the sample still included multiple ethnic backgrounds, allowing for reasonably diverse perspectives.

The age distribution was skewed toward younger adults, with a majority of participants aged 25–34 years (57%), followed by 35–44 years (25%) and 45–59 years (11%). Smaller proportions of the sample were 19–24 years (3%), 60 years and above (3%), and under 19 years (1%), suggesting that the findings may be most representative of young to middle-aged adults.

Educational attainment among participants was also varied. Most participants held a Bachelor's degree (68%), accompanied by those with occupational or vocational training (12%), graduate-level degrees (10%), high school or equivalent (9%), and below high school education (1%). Overall, this indicates a well-educated sample with a range of educational experiences, which supports the generalizability of the results across different educational backgrounds. The details of our participants' demographics are reported in Figure 1.

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Appendix B: Questionnaire of our human study

Part I. Questions about sharing the displayed news story

Suppose you see the above news online, please answer the following questions.

Q1. Would you share this piece of news with your friends on social media?
- Yes
- No

Q2. If yes, please share with us your reasons for sharing this news on social media. (Select all that apply.)
- It will generate discussion among my friends.
- I want to share this news because it is relevant to my life.
- I want to appear good.
- I want to appear smart.
- I want to share this news because I believe it is true.
- I want to share this news because I believe the headline is true although I am not sure about the supporting image/video.
- I want to share this news because I believe the supporting image/video is true although I am not sure about the headline.
- I want to share this news because I agree with it.
- Others. Please specify: _____

Q3. If no, please share with us your reasons for NOT to share this news on social media. (Select all that apply.)
- I do not want this news on my social media page because it is not interesting.
- I will not generate discussion among my friends.

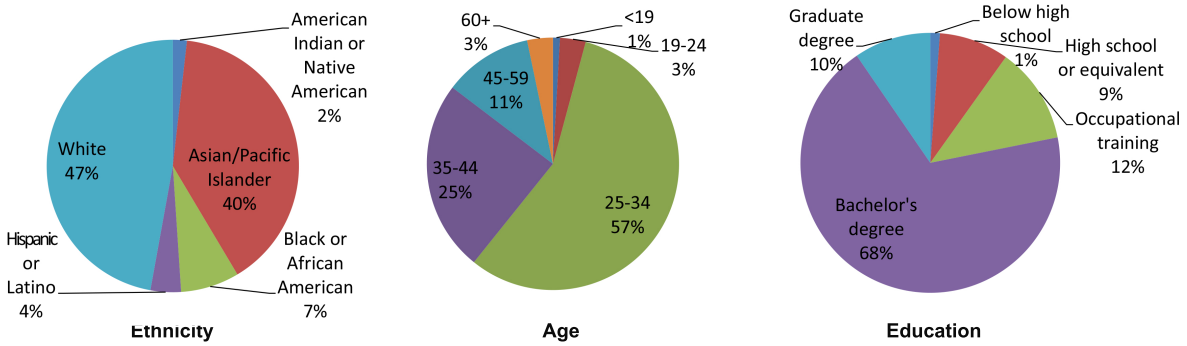

FIGURE 1. Demographics statistics and social media usage of the 272 participants in our human studies.

- I do not want to share this news because it is not relevant to my life.
- I do not want to appear bad.
- I do not want to appear stupid.
- I do not want to share this news even without the red warning text, because I believe the news headline is not true.
- I do not want to share this news even without the red warning text, because I believe the supporting image/video is not true.
- I do not want to share this news even without the red warning text, because I do not agree with it.
- I am influenced by the red warning text regarding the credibility of the news.
- Others. Please specify: _____

Q4. Please tell us whether and how you utilized the red warning text regarding the credibility of the video (when such information was available).

- It has no influence to me on whether to share this news.
- It has some influence to me on whether to share this news.
- It has great influence to me on whether to share this news.

Q5. There is no red warning text for this news.

Q6. When judging the credibility of a piece of news, how important would you consider each of the following pieces of information? (On a scale of 1-5, with 1 being Not at all important and 5 being Extremely important)

Q1-1. A majority of social media users dispute the credibility of this news.

Q2-2. Multiple fact-checking journalists dispute the credibility of this news.

Q3-3. Major news outlets dispute the credibility of this news.

Q4-4. Computer algorithms using Artificial Intelligence (AI) techniques dispute the credibility of this news.

Q7. When deciding whether or not to share the news, how helpful is each of the following pieces of information on your decision? (On a scale of 1-5, with 1 being Not at all helpful and 5 being Extremely helpful)

Q1-1. A majority of social media users dispute the credibility of this news.

Q2-2. Multiple fact-checking journalists dispute the credibility of this news.

Q3-3. Major news outlets dispute the credibility of this news.

Q4-4. Computer algorithms using Artificial Intelligence (AI) techniques dispute the credibility of this news.

Part II. Basic demographics

The following background survey helps us understand your choice better.

Q1. Your gender:

- Male
- Female
- Others

Q2. Your race:

- American Indian or Native American
- Asian / Pacific Islander
- Black or African American
- Hispanic or Latino
- White
- Other

Q3. Your age:

- <19
- 19-24
- 25-34
- 35-44
- 45-59

- 60+

Q4. Your education:
- Below high school
- High school graduate (includes equivalency)
- Trade/technical/vocational training
- Bachelor's degree
- Master's degree (e.g., M.S. and Ph.D.)
- Graduate degree (e.g., M.S. and Ph.D.)

Q5. In what country do you currently live?
- United States
- India
- Canada
- United Kingdom
- Other European countries
- Other Asian countries
- Oceania countries
- None of the above

Part III. Habit of using social media

Considering your complete experience with social media websites (Facebook, Twitter, Instagram, etc.), how likely are you to recommend it to your family and friends?

Q1. How many years have you used social media?
- 1
- 2
- 3
- 4
- 5
- More than 5

Q2. Please indicate the frequency of using each social media platform? (Select all that apply.)
- Facebook
- Twitter
- Instagram
- Snapchat
- Google+
- LinkedIn
- Wechat
- Weibo
- Tumblr
- Pinterest
- Other. Please specify: _____

Q3. How many friends do you have on social media?
- Less than 5
- 5-20
- 21-40
- 41-60
- 61-80
- 81-100
- More than 100

Q4. How many friends from your social media website do you know in person?
- All of them
- Most of them
- About half of them
- A few of them
- None of them

Q5. How often do you share content with your friends on social media?
- Never
- Every few months
- Every few weeks
- Weekly
- Daily
- Multiple times a day

Q6. How often do you use the chat app on your social media accounts?
- Not at all often
- Slightly often
- Moderately often
- Very often
- Extremely often

Q7. On average, how long do you spend each day reading, watching, or listening to the news on social media?
- Less than 30 mins
- An hour
- 1-2 hours
- 2-4 hours
- More than 4 hours
- Never

Q8. What is your purpose of using social media website? (Select all that apply.)
- To keep in touch with friends and family
- To make friends
- To socialize casually
- Entertainment
- Educational
- Buying and selling
- Travel related
- Event planning
- To find employment
- For news and current events
- To gather information happening in town
- Dating/flirting
- To browse/waste time
- Other. Please specify: _____

Q9. When do you access social media websites? (Select all that apply.)
- Traveling/commuting
- While at school/university/work

- During social occasions
- During meal times
- Social gathering/party
- Dating
- Holiday
- Before going to bed

Q10. Do you turn on audio function while browsing videos on social media (e.g., when you open Instagram or Facebook, videos will autoplay with the sound off, will you turn on the sound)?

Q11. Which content do you share with your friends on social media, if any? (Select all that apply.)

- External links (content generated by others)
- Own content (content generated by myself)
- Pictures/videos
- Livestream videos
- Status posts without pictures/videos
- News headlines/articles
- Blogs
- Comments on others' posts
- Other. Please specify: _____

Q12. Why do you share content with your friends on social media? (Select all that apply.)

- I want my social media presence to be interesting to my friends on social media.
- I want to generate discussion among my friends on social media.
- I want to share humorous content.
- I want to look good.
- I want to look smart.
- I want to share content of relevance to my life.
- I want to share content of relevance to the lives of my friends on social media.
- I want to share informative and educational content.
- I want to respond/react to the content posted by my friends on social media.
- Other. Please specify: _____

Part III. Relating to News

In general, how interested are you in news?

- Extremely interested
- Very interested
- Moderately interested
- Slightly interested
- Not interested at all

Which of the following types of news are important to you? (Select all that apply.)

- News about the Country's government
- Local News about my town or city
- News about my region
- Business and financial news
- News about the economy
- Entertainment and celebrity news
- Fun and/or weird news
- Health and education news
- Arts and culture news
- Sports news
- News about politics
- Science and technology news
- Other. Please specify: _____

Which of the following do you consider the most important source for news and information?

- Print newspapers
- Radio
- Local television
- National television (e.g., ABC, BBC, Dangal TV)
- Cable television (e.g., CNN, Fox News, MTV, StarPlus)
- Online news Web sites or Apps excluding social media and blogs
- Blogs
- Social media (e.g., Facebook, Twitter)
- Other. Please specify: _____

How often do you share news with your friends on social media?

- Never
- Less than once a week
- Once a week
- 2-3 times a week
- 4-6 times a week
- Daily

How concerned are you about privacy breach news (e.g., the Facebook-Cambridge Analytica data scandal)?

Has your concern about fake news affected your behavior on social media?

- Yes it greatly affected my behavior on social media
- Yes it slightly affected my behavior on social media
- No it does not affect my behavior on social media

If you choose yes in the above question, please specify how it changed your behavior.

- I share news less frequently
- I share news to fewer number of friends
- I think more carefully before sharing the news
- I will double check the credibility of the news before sharing it.
- It does not affect my behavior.
- Others, please specify: _____